Part 1: Phase-Wise Deployment Roadmap

Phase	Phase Focus & Milestone	Targeted Modules	Total Components
Phase 0	**Inception & Mobilisation**	Dev/Test infra, State Data Centre approvals, baseline architecture	Foundation
Phase 1	**Foundation & Revenue Core**	**PTX** (33), **WTR** (32), **GRV** (30), **FIN** (30), **DMS** (30)	**155 Components**
Phase 2	**Regulation & Licensing**	**TRL** (26), **UCH** (23), **BPM** (22), **TPL** (23), **FNC** (17)	**111 Components**
Phase 3	**Field Operations & Infrastructure**	**SWM** (17), **ENG** (20), **SLM** (12), **PRK** (18), **INV** (17)	**84 Components**
Phase 4	**Administration, Estates & Welfare**	**HRM** (19), **MPM** (15), **LGL** (17), **GSC** (14)	**65 Components**
TOTAL	**Full SaaS Enterprise Platform**	**19 Modules**	**415 Components**

Part 2: Complete Catalogue of All 415 Components

Phase 1: Foundation & Revenue Core (155 Components)

1. [PTX] Property Tax & Revenue Management (33 Components)

- `PTX-P01` : Property registration and identification
- `PTX-P02` : Legacy data onboarding and state platform sync
- `PTX-P03` : Property survey and GIS reconciliation
- `PTX-P04` : Property data correction
- `PTX-P05` : New assessment – built property
- `PTX-P06` : Self-assessment by owner
- `PTX-P07` : Apartment / multi-unit assessment
- `PTX-P08` : Vacant land tax
- `PTX-P09` : Reassessment – addition, alteration, usage or occupancy change
- `PTX-P10` : General revision of rates and zones
- `PTX-P11` : Back-assessment of unassessed or under-assessed property
- `PTX-P12` : Setback verification and deviation penalty
- `PTX-P13` : Exemptions, concessions and category rebates
- `PTX-P14` : Auto-mutation from registration
- `PTX-P15` : Mutation by application
- `PTX-P16` : Bifurcation and amalgamation
- `PTX-P17` : Demolition and deletion
- `PTX-P18` : Periodic demand generation (half-yearly)
- `PTX-P19` : Bill amendment and demand correction
- `PTX-P20` : Bill distribution and notifications
- `PTX-P21` : Collection (online, counter, POS, BBPS)
- `PTX-P22` : Cheque bounce, receipt cancellation and refund
- `PTX-P23` : Daily reconciliation and remittance

- PTX–P24 : Objection, revision petition and appeal
- PTX–P25 : Interest, penalty and recovery
- PTX–P26 : One-time settlement and interest waiver
- PTX–P27 : Write-off and remission
- PTX–P28 : Property tax certificates (NOC, dues clear)
- PTX–P29 : Property tax complaints
- PTX–P30 : Dashboards, DCB and compliance reporting
- PTX–P31 : Part-area non-residential use and rooftop installations
- PTX–P32 : Road and location classification for valuation
- PTX–P33 : Self-declared commercial accommodation use

2. [WTR] Water, Sewerage & Utility Billing (32 Components)

- WTR–W01 : New water connection
- WTR–W02 : New sewerage / UGD connection
- WTR–W03 : Regularisation of unauthorised connection
- WTR–W04 : Bulk / apartment connection with sub-metering
- WTR–W05 : Temporary connection
- WTR–W06 : Ownership transfer
- WTR–W07 : Usage or category change
- WTR–W08 : Disconnection
- WTR–W09 : Reconnection
- WTR–W10 : Connection surrender / permanent closure
- WTR–W11 : Meter installation, replacement and calibration
- WTR–W12 : Meter reading and spot billing
- WTR–W13 : Estimated and average billing
- WTR–W14 : Unmetered flat-rate billing
- WTR–W15 : Sewerage charge calculation
- WTR–W16 : Bill generation, adjustment and dispatch
- WTR–W17 : Collection (counters, digital, BBPS, mPOS)
- WTR–W18 : Advance payment and deposit ledger
- WTR–W19 : Dues recovery, interest and penalty
- WTR–W20 : Defaulter management and red-notice workflow
- WTR–W21 : Water supply schedule and pressure monitoring
- WTR–W22 : Water quality testing and potability register
- WTR–W23 : Bulk water management and NRW estimation
- WTR–W24 : Pipeline maintenance and breakdown logging
- WTR–W25 : UGD overflow and blockage clearance
- WTR–W26 : STP / FSTP intake and discharge monitoring
- WTR–W27 : Water tanker booking and delivery tracking
- WTR–W28 : Consumer grievance redressing
- WTR–W29 : Field inspection and tap testing
- WTR–W30 : Consumer portal and WhatsApp self-service
- WTR–W31 : Water & sewerage analytics and DCB reporting
- WTR–W32 : Rainwater harvesting compliance verification

3. [GRV] Citizen Service Desk, Grievance & RTI (30 Components)

- GRV–G01 : Omnichannel grievance intake
- GRV–G02 : Citizen identity and contact verification
- GRV–G03 : Auto-routing and ticket classification

- GRV-G04 : WhatsApp conversational grievance assistant
- GRV-G05 : Integration with Swachhata / state citizen platforms
- GRV-G06 : Location geotagging and ward resolution
- GRV-G07 : Duplicate detection and ticket merging
- GRV-G08 : Priority and VIP grievance assignment
- GRV-G09 : SLA calculation and countdown tracking
- GRV-G10 : Field staff work assignment and dispatch
- GRV-G11 : Mobile field resolution and photo evidence capture
- GRV-G12 : Internal transfer and inter-department referral
- GRV-G13 : Work extension request and approval
- GRV-G14 : Reopening of unresolved grievances
- GRV-G15 : Automated escalation and alerts
- GRV-G16 : Quality audit of resolved complaints
- GRV-G17 : Citizen feedback and satisfaction rating
- GRV-G18 : Call centre agent console and outbound IVRS
- GRV-G19 : Bulk notification and ward announcement broadcasts
- GRV-G20 : RTI application intake and fee payment
- GRV-G21 : RTI routing to PIO / APIO
- GRV-G22 : RTI response preparation and document upload
- GRV-G23 : RTI first appeal and appellate authority workflow
- GRV-G24 : RTI second appeal (State Information Commission sync)
- GRV-G25 : Proactive disclosure register (Section 4 RTI)
- GRV-G26 : Grievance and RTI dashboards and heatmap
- GRV-G27 : Ward committee and councillor grievance review
- GRV-G28 : SLA compliance reporting for commissioner
- GRV-G29 : Chronic complaint hotspot analysis
- GRV-G30 : Citizen happiness index tracking

4. [FIN] Finance & Accounting (30 Components)

- FIN-F01 : Chart of accounts and NMAM compliance
- FIN-F02 : Financial year opening and period lock
- FIN-F03 : Budget preparation, estimates and BE/RE compilation
- FIN-F04 : Budget allocation, allotment and re-appropriation
- FIN-F05 : Budgetary control and commitment tracking
- FIN-F06 : Double-entry journal voucher processing
- FIN-F07 : Automated revenue collection journals
- FIN-F08 : Vendor and contractor bill passing
- FIN-F09 : Payment order and multi-signatory authorisation
- FIN-F10 : Electronic payment disbursement (e-Payment / PFMS)
- FIN-F11 : Bank account master and treasury liaison
- FIN-F12 : Automated bank reconciliation statement (BRS)
- FIN-F13 : Petty cash and imprest account management
- FIN-F14 : Inter-bank fund transfers and investment register
- FIN-F15 : Grants management (State Finance Commission, 15th FC)
- FIN-F16 : Scheme-wise fund tracking and utilisation certificates (UC)
- FIN-F17 : Municipal loan and debt servicing register
- FIN-F18 : TDS and tax deduction at source processing
- FIN-F19 : GST filing, input tax credit and return generation
- FIN-F20 : Labour cess and statutory deduction accounting

- `FIN-F21` : Fixed asset capitalisation and depreciation
- `FIN-F22` : General ledger, trial balance and cash book
- `FIN-F23` : Receipt and payment account generation
- `FIN-F24` : Income and expenditure statement
- `FIN-F25` : Balance sheet generation
- `FIN-F26` : Internal audit queries and para management
- `FIN-F27` : CAG / Local Fund Audit (LFA) compliance tracking
- `FIN-F28` : Real-time financial dashboard and liquidity monitor
- `FIN-F29` : Municipal bonds and credit rating tracking
- `FIN-F30` : Year-end closing and financial statement publishing

5. [DMS] Document Management System (30 Components)

- `DMS-D01` : Document intake from ERP modules
- `DMS-D02` : Direct document upload and scanning
- `DMS-D03` : Optical character recognition (OCR) and text extraction
- `DMS-D04` : Automated classification and metadata tagging
- `DMS-D05` : Barcode and QR code generation for physical files
- `DMS-D06` : Integration with DigiLocker for document fetching
- `DMS-D07` : Property dossier compilation and chain-of-title linking
- `DMS-D08` : Version control, revision tracking and audit trail
- `DMS-D09` : Electronic signing (Aadhaar eSign, DSC USB token)
- `DMS-D10` : Document stamping, watermarking and redaction
- `DMS-D11` : E-noting and green sheet processing
- `DMS-D12` : File movement and dispatch register
- `DMS-D13` : Document access control and confidentiality levels
- `DMS-D14` : Full-text search and faceted document retrieval
- `DMS-D15` : Document expiry, renewal alert and retention policies
- `DMS-D16` : Archival and record room indexation
- `DMS-D17` : Document disposal and shredding compliance
- `DMS-D18` : Physical document tracking (RFID / barcode in record room)
- `DMS-D19` : Bulk document export for audits and courts
- `DMS-D20` : Document sharing via secure time-limited links
- `DMS-D21` : Template management for official orders and NOCs
- `DMS-D22` : Gazette notification repository
- `DMS-D23` : Municipal council resolutions and minutes archive
- `DMS-D24` : Legacy record digitisation tracking
- `DMS-D25` : Public records portal (citizen document access)
- `DMS-D26` : RTI document bundle preparation
- `DMS-D27` : Storage quota and deduplication management
- `DMS-D28` : Document access logs and audit reporting
- `DMS-D29` : Integrity verification and tamper detection (Blockchain anchoring)
- `DMS-D30` : Central document dashboard and metrics

Phase 2: Regulation & Licensing (111 Components)

6. [TRL] Trade Licence Management (26 Components)

- `TRL-T01` : New trade licence application

- TRL–T02 : Trade classification and schedule-of-fees mapping
- TRL–T03 : Dangerous and offensive (D&O) trade scrutiny
- TRL–T04 : Co-living, hostel and paying-guest licensing
- TRL–T05 : Factory and industrial establishment licence
- TRL–T06 : Food and beverage establishment scrutiny
- TRL–T07 : Trade licence renewal (auto and scrutiny-based)
- TRL–T08 : Trade amendment (name, ownership, location, area)
- TRL–T09 : Trade licence cancellation and suspension
- TRL–T10 : Trade closure and surrender
- TRL–T11 : Site inspection and verification by health inspector
- TRL–T12 : Document scrutiny and NOC validation
- TRL–T13 : Trade licence fee calculation and penalties
- TRL–T14 : Provisional trade licence issuance
- TRL–T15 : Final licence generation with QR verification
- TRL–T16 : Inter-module Fire NOC status verification
- TRL–T17 : Property tax dues clearance validation
- TRL–T18 : Trade licence certificate reprint and duplicate
- TRL–T19 : Unlicensed trade survey and identification
- TRL–T20 : Show-cause notice and enforcement action
- TRL–T21 : Trade seizure and sealing workflow
- TRL–T22 : Street vending zone demarcation and survey
- TRL–T23 : Street vendor ID card and certificate of vending
- TRL–T24 : Trade licence demand, collection and balance (DCB)
- TRL–T25 : WhatsApp and SMS renewal reminders
- TRL–T26 : Trade licensing dashboards and revenue forecasting

7. [UCH] User Charges Management (23 Components)

- UCH–U01 : Automated user charge registration from property data
- UCH–U02 : Bulk waste generator (BWG) charge setup
- UCH–U03 : Commercial waste collection fee assessment
- UCH–U04 : Septage and sewage collection charges
- UCH–U05 : Trade waste and special bio-waste surcharge
- UCH–U06 : Street cutting and road restoration fee
- UCH–U07 : Combined utility billing with property / water
- UCH–U08 : Standalone user charge demand generation
- UCH–U09 : Spot fines and sanitation violation penalties
- UCH–U10 : User charge collection (POS, digital, counter)
- UCH–U11 : Advance user charge payment ledger
- UCH–U12 : Defaulter escalation and compounding interest
- UCH–U13 : Exemption and concession management
- UCH–U14 : Field sanitation inspector collection sync
- UCH–U15 : Direct debit and recurring mandate processing
- UCH–U16 : User charge revision and notification workflow
- UCH–U17 : Reconciliation and bank remittance
- UCH–U18 : Citizen protest and charge recalculation
- UCH–U19 : User charge demand-collection-balance (DCB)
- UCH–U20 : Mobile notification and invoice delivery
- UCH–U21 : Public parking fees and contractor collection
- UCH–U22 : Hoarding, banner and advertisement display fees

- `UCH-U23` : Comprehensive user charges analytics dashboard

8. [BPM] Building Permissions (22 Components)

- `BPM-B01` : Building permission application intake (TG-bPASS mode)
- `BPM-B02` : Instant approval / self-certification for small plots
- `BPM-B03` : Single-window clearance for medium / large structures
- `BPM-B04` : Pre-construction document validation
- `BPM-B05` : Auto-DCR CAD drawing scrutiny and rule check
- `BPM-B06` : Master plan zoning and land use verification
- `BPM-B07` : Multi-department NOC coordination
- `BPM-B08` : Site inspection scheduling and geo-tagged reports
- `BPM-B09` : Fee calculation, betterment charges and mortgage deed
- `BPM-B10` : Building permit issuance with security barcode
- `BPM-B11` : Permission revocation, cancellation and modification
- `BPM-B12` : Construction stage inspection (plinth, roof level)
- `BPM-B13` : Work commencement notice and inspection logging
- `BPM-B14` : Deviation detection and compounding penalty
- `BPM-B15` : Occupancy certificate (OC) application and issuance
- `BPM-B16` : Partial occupancy certificate
- `BPM-B17` : Demolition permission and debris disposal plan
- `BPM-B18` : Unauthorised construction detection and show cause
- `BPM-B19` : Building sealing and demolition enforcement workflow
- `BPM-B20` : Architect / engineer registration and blacklisting
- `BPM-B21` : Public building register and status transparency
- `BPM-B22` : Building permission analytics and revenue tracking

9. [TPL] Town Planning (23 Components)

- `TPL-P01` : Master plan GIS layer management and zone lookups
- `TPL-P02` : Change of land use (CLU) application and processing
- `TPL-P03` : Zoning certificate issuance
- `TPL-P04` : Land acquisition and compensation tracking
- `TPL-P05` : Alignment drawing and road widening reservations
- `TPL-P06` : Layout permission application intake
- `TPL-P07` : Layout scrutiny and technical sanction
- `TPL-P08` : Open space and road dedication deeds
- `TPL-P09` : Final layout approval and plot release
- `TPL-P10` : Gated community and township layout clearance
- `TPL-P11` : Layout regularisation scheme (LRS) intake
- `TPL-P12` : LRS scrutiny, penalty calculation and approval
- `TPL-P13` : Unapproved layout detection and red-flagging
- `TPL-P14` : Transferable development rights (TDR) issuance
- `TPL-P15` : TDR bank, ledger and trade validation
- `TPL-P16` : TDR utilisation against building permission
- `TPL-P17` : Heritage structure and conservation zone monitoring
- `TPL-P18` : Lake / waterbody buffer zone enforcement
- `TPL-P19` : Encroachment reporting and field removal tracking
- `TPL-P20` : Town planning notices and orders repository
- `TPL-P21` : Town planning committee resolution records
- `TPL-P22` : Public scrutiny and objections register

- `TPL-P23` : Town planning spatial analytics and growth forecasting

10. [FNC] Fire NOC Management (17 Components)

- `FNC-N01` : Integration with State Disaster Response and Fire Service
- `FNC-N02` : Fire safety classification and risk grading
- `FNC-N03` : Provisional Fire NOC application for buildings
- `FNC-N04` : Final Occupancy Fire NOC processing
- `FNC-N05` : Fire NOC renewal application
- `FNC-N06` : Event and temporary structure fire safety clearance
- `FNC-N07` : Fire safety checklist verification and equipment audit
- `FNC-N08` : Joint inspection with fire department officers
- `FNC-N09` : Fire safety fee and penalty calculation
- `FNC-N10` : Fire NOC digital certificate issuance with QR
- `FNC-N11` : Automatic periodic fire audit reminders
- `FNC-N12` : Non-compliance notices and evacuation alerts
- `FNC-N13` : Fire NOC revocation and building sealing trigger
- `FNC-N14` : Commercial hazardous material storage verification
- `FNC-N15` : GIS fire risk mapping and hydrant proximity view
- `FNC-N16` : Fire incident reporting and post-disaster audit
- `FNC-N17` : Fire safety compliance dashboard and metrics

Phase 3: Field Operations & Infrastructure (84 Components)

11. [SWM] Solid Waste Management (17 Components)

- `SWM-S01` : Waste generator categorization and master register
- `SWM-S02` : Bulk waste generator (BWG) on-site processing audit
- `SWM-S03` : Door-to-door collection tracking and RFID logging
- `SWM-S04` : Sanitation worker attendance and route allocation
- `SWM-S05` : Secondary collection and community bin monitoring
- `SWM-S06` : Fleet tracking, route optimisation and GPS playback
- `SWM-S07` : Transfer station intake and weighbridge integration
- `SWM-S08` : Processing facility / compost plant monitoring
- `SWM-S09` : Construction and demolition (C&D) waste pickup
- `SWM-S10` : Hazardous, electronic and sanitary waste logging
- `SWM-S11` : Dump yard / landfill bio-mining and capping tracking
- `SWM-S12` : Blackspot identification and clearance verification
- `SWM-S13` : Swachh Auto Tipper (SAT) maintenance and fuel log
- `SWM-S14` : Ragpicker and informal waste worker formalisation
- `SWM-S15` : Swachh Survekshan indicator tracking and audit
- `SWM-S16` : Citizen waste reporting (overflowing bin, burning)
- `SWM-S17` : Solid waste management operations dashboard

12. [ENG] Engineering Projects & e-MB (20 Components)

- `ENG-E01` : Works demand generation and council proposal
- `ENG-E02` : Administrative sanction (AS) workflow
- `ENG-E03` : Detailed project report (DPR) and estimate preparation
- `ENG-E04` : Technical sanction (TS) with standard schedule of rates

- `ENG-E05` : Tender preparation and e-Procurement sync
- `ENG-E06` : Work order issuance and contractor assignment
- `ENG-E07` : Milestone planning and project scheduling
- `ENG-E08` : Electronic Measurement Book (e-MB) entry
- `ENG-E09` : Quality control (QC) inspection and test certificate
- `ENG-E10` : Contractor work bill generation from e-MB
- `ENG-E11` : Bill scrutiny, statutory deduction and passing
- `ENG-E12` : Running account (RA) and final bill settlement
- `ENG-E13` : Defect liability period (DLP) and retention release
- `ENG-E14` : Work completion certificate issuance
- `ENG-E15` : GIS project geotagging and progress photo logging
- `ENG-E16` : Road asset register and trenching permission
- `ENG-E17` : Contractor performance rating and blacklisting
- `ENG-E18` : Cost escalation and time extension approvals
- `ENG-E19` : Works audit queries and vigilance enquiry tracking
- `ENG-E20` : Engineering projects executive dashboard

13. [SLM] Street Light Management (12 Components)

- `SLM-L01` : Street light pole and fixture GIS asset inventory
- `SLM-L02` : Centralised Control and Monitoring System (CCMS) sync
- `SLM-L03` : Fault detection and citizen outage complaint intake
- `SLM-L04` : Auto-dispatch to electrical maintenance contractor
- `SLM-L05` : Field repair verification and before/after photo proof
- `SLM-L06` : Automated switching schedules and dimmer controls
- `SLM-L07` : Burning hours analysis and energy consumption tracking
- `SLM-L08` : Electricity DISCOM meter reading and bill audit
- `SLM-L09` : LED conversion and energy performance contracting
- `SLM-L10` : Preventative maintenance schedule and replacement log
- `SLM-L11` : Contractor SLA and penalty deduction calculator
- `SLM-L12` : Street light operations dashboard and uptime tracker

14. [PRK] Parks & Horticulture Management (18 Components)

- `PRK-K01` : Municipal park and open space GIS asset register
- `PRK-K02` : Urban tree census and QR-tagging register
- `PRK-K03` : Park maintenance contract management and inspections
- `PRK-K04` : Playground equipment and asset safety check
- `PRK-K05` : Nursery management and plant sapling inventory
- `PRK-K06` : Green belt and central median plantation log
- `PRK-K07` : Haritha Haram / urban greening campaign tracking
- `PRK-K08` : Water supply and sprinkler automation monitoring
- `PRK-K09` : Composting of garden waste in parks
- `PRK-K10` : Park entry ticketing and visitor pass issuance
- `PRK-K11` : Booking of parks / open spaces for public events
- `PRK-K12` : Commercial food kiosk and parking leases in parks
- `PRK-K13` : Vandalism and damage reporting in parks
- `PRK-K14` : Corporate social responsibility (CSR) adoption of parks
- `PRK-K15` : Tree felling / pruning permission and inspection
- `PRK-K16` : Compensatory afforestation tracking and compliance
- `PRK-K17` : Biodiversity register and fauna documentation

- `PRK-K18` : Parks & horticulture dashboard and canopy coverage

15. [INV] Inventory & Stores (17 Components)

- `INV-V01` : Master item catalogue and classification
- `INV-V02` : Central store and divisional warehouse register
- `INV-V03` : Stock level monitoring and reorder point alerts
- `INV-V04` : Departmental indent intake and approval
- `INV-V05` : Purchase order (PO) generation and vendor linkage
- `INV-V06` : Goods receipt note (GRN) and gate pass logging
- `INV-V07` : Material inspection and quality acceptance
- `INV-V08` : Stock issuance and transfer between departmental stores
- `INV-V09` : Stock return and repair tracking
- `INV-V10` : Scrap, condemned item and auction management
- `INV-V11` : Fuel inventory and vehicle refuelling slip log
- `INV-V12` : Annual physical stock verification and reconciliation
- `INV-V13` : Stock valuation methods (FIFO / Weighted Average)
- `INV-V14` : Emergency procurement and disaster stockpile
- `INV-V15` : Vendor delivery performance and lead time analysis
- `INV-V16` : Integration with Finance module for store ledger
- `INV-V17` : Central stores executive dashboard and stock ledger

Phase 4: Administration, Estates & Welfare (65 Components)

16. [HRM] Payroll & HR Management (19 Components)

- `HRM-H01` : Sanctioned post and establishment schedule
- `HRM-H02` : Electronic Service Register (e-SR) for municipal staff
- `HRM-H03` : Employee onboarding, deputation and contract staff
- `HRM-H04` : Transfer and posting order processing
- `HRM-H05` : Biometric / geo-fenced attendance tracking
- `HRM-H06` : Leave application, encashment and surrender
- `HRM-H07` : Tour, travel allowance (TA/DA) and advance
- `HRM-H08` : Medical reimbursement claim processing
- `HRM-H09` : Monthly payroll processing and pay-bill passing
- `HRM-H10` : Statutory deductions (CPS/NPS, GPF, PT, IT TDS)
- `HRM-H11` : Festival and special advance disbursement
- `HRM-H12` : Annual increments and promotion workflows
- `HRM-H13` : Disciplinary proceedings and charge memo tracking
- `HRM-H14` : Performance appraisal (PAR / APAR) workflow
- `HRM-H15` : Staff training, capacity building and certifications
- `HRM-H16` : Retirement, superannuation and pension processing
- `HRM-H17` : Gratuity and terminal benefits calculation
- `HRM-H18` : Outsource labour agency billing and compliance
- `HRM-H19` : HR & payroll executive dashboard and headcount stats

17. [MPM] Municipal Property Management (15 Components)

- `MPM-M01` : Municipal immovable property and land register
- `MPM-M02` : GIS geotagging and boundary demarcation of lands

- `MPM-M03` : Title deed, revenue survey number and patta vault
- `MPM-M04` : Vacant municipal land bank and future use zoning
- `MPM-M05` : Municipal commercial complex and market stall master
- `MPM-M06` : Auction, lease allocation and tenancy agreement
- `MPM-M07` : Lease rent periodic demand and escalation
- `MPM-M08` : Rent collection, receipt generation and arrears
- `MPM-M09` : Lease transfer, subletting and tenancy renewal
- `MPM-M10` : Eviction of overstayed tenants and recovery notices
- `MPM-M11` : Encroachment alert, drone surveillance and removal
- `MPM-M12` : Municipal asset valuation and insurance portfolio
- `MPM-M13` : Maintenance and civil works expenditure tracking
- `MPM-M14` : Revenue sharing from hoardings, towers and ATMs
- `MPM-M15` : Municipal property revenue and asset dashboard

18. [LGL] Legal Case Management (17 Components)

- `LGL-J01` : Court case registration and litigation tracking
- `LGL-J02` : eCourts / High Court CNR number synchronization
- `LGL-J03` : Departmental assignment and Standing Counsel mapping
- `LGL-J04` : Notice, summon and writ petition document upload
- `LGL-J05` : Counter affidavit and parawise remark drafting
- `LGL-J06` : Hearing date calendar, cause list and diary
- `LGL-J07` : Limitation period calculation and urgency alerts
- `LGL-J08` : Interim stay order and injunction tracking
- `LGL-J09` : Contempt of court warning and priority escalations
- `LGL-J10` : Judgement and final decree repository
- `LGL-J11` : Appeal filing recommendation and deadline tracker
- `LGL-J12` : Legal fee bill verification and payment sanction
- `LGL-J13` : Out-of-court settlement and Lok Adalat matters
- `LGL-J14` : Departmental enquiry and legal opinion requests
- `LGL-J15` : Advocate performance monitoring and empanelment
- `LGL-J16` : Municipal law, rule and bylaw master database
- `LGL-J17` : Legal risk and litigation summary dashboard

19. [GSC] Government Schemes (14 Components)

- `GSC-C01` : Central and State welfare scheme catalogue
- `GSC-C02` : Beneficiary survey, application intake and KYC
- `GSC-C03` : Socio-economic eligibility scoring and validation
- `GSC-C04` : PMAY-Urban housing application and stage tracking
- `GSC-C05` : Geo-tagged construction stage verification for housing
- `GSC-C06` : Housing allotment lottery and patta certificate
- `GSC-C07` : PM SVANidhi street vendor micro-credit loan link
- `GSC-C08` : Self-Help Group (SHG / MEPMA) group registration
- `GSC-C09` : SHG bank linkage and revolving fund tracking
- `GSC-C10` : Direct Benefit Transfer (DBT) sanction and mandate
- `GSC-C11` : Scheme utilisation certificate (UC) generation
- `GSC-C12` : Beneficiary grievance and assistance desk
- `GSC-C13` : Social audit and third-party verification reports
- `GSC-C14` : Welfare schemes executive dashboard and coverage heatmap

Verification Summary

- **Total Modules**: 19
- **Total Granular RBAC Components**: 415
- **Sum by Phase**: Phase 1 (155) + Phase 2 (111) + Phase 3 (84) + Phase 4 (65) = **415 Total**
- **Catalogued Location**: docs/MODULE_COMPONENTS_CATALOGUE.md